Course program and reading list

Semester 2 Year 2023

School: Efi Arazi School of Computer Science M.Sc.

Natural Language Processing

Lecturers:

Mr. Amir Cohen Amir.Cohen02@post.runi.ac.il

Teaching Assistants:

Mr. Sahar Millis sahar.millis@post.runi.ac.il

Mr. Royi Rassin royi.rassin@post.runi.ac.il

Course No.:	Course Type :	Weekly Hours :	Credit:
3523	Elective	3	3

Course Requirements :	Group Code :	Language:
Final Paper	232352302	English

Prerequisites

Prerequisite:

52 - Calculus I
53 - Calculus II
54 - Linear Algebra I
55 - Linear Algebra II
56 - Discrete Mathematics
59 - Data Structures
69 - Logic and Set Theory
77 - Algorithms
417 - Introduction to Computer Science
3141 - Machine Learning from Data

Course Description

his course offers an introduction to Natural Language Processing (NLP) techniques and their applications. Students will gain a solid understanding of the fundamental concepts and methods used in NLP, along with hands-on experience in implementing algorithms and building applications. The course will cover topics such as text processing, syntactic and semantic analysis, machine translation, sentiment analysis, and dialogue systems. Students will also be exposed to the latest advancements in NLP, including deep learning techniques and neural networks.*

Section 1: Introduction to NLP and ChatGPT

- Overview of NLP
- Linguistic foundations
- Applications of NLP
- Introduction to ChatGPT and GPT architectures

Section 2: Language Models (LMs)

- N-gram models
- Lm evaluation
- Smoothing

Section 3: Part-of-Speech Tagging

- Rule-based tagging
- Probabilistic tagging
- Machine learning approaches for tagging

Section 4: Introduction to Deep Learning (DL) for NLP

- Feedforward neural networks
- Convolutional neural networks
- Activation functions and optimization techniques

Section 5: Word Embeddings

- One-hot encoding
- Word2Vec, GloVe, and FastText
- Contextual embeddings (ELMo, BERT)

Section 6: Long Short-Term Memory (LSTM) Networks

- Recurrent neural networks (RNNs)
- LSTM and GRU architectures
- Applications in NLP

Section 7: Attention Mechanisms

- The attention mechanism concept
- Self-attention and multi-head attention
- Applications in NLP

Section 8: Transformers

- Transformer architecture
- BERT, GPT, and other variants
- Fine-tuning for specific tasks

Section 9: Downstream Task - Question Answering (QA)

- Factoid QA
- Reading comprehension
- Retrieval based QA

Section 10: Downstream Task - Summarization and Translation

- Extractive and abstractive summarization
- Neural machine translation
- Techniques for improving quality and efficiency

Section 11: Final Lecture and Guest Lecture

- Recap of course content
- Trends and future directions in NLP research
- Guest lecture: Expert insights into real-world NLP applications and challenges

*This text was generated with the help of chatGPT

Course Goals

This course aims to introduce undergraduate and postgraduate CS students to research and development in computational linguistics and natural language processing. The course covers linguistic, algorithmic and statistical essentials, and demonstrates these through solutions to real-world challenges.

Grading

- **Grades**

50% assignments

50% final project

- **Assignments**

4 assignments in total. The wight of the later assignments will be higher.

Ex1 - 10%

Ex2 - 10%

Ex3 - 15%

Ex4 - 15%

- **Competition**

We will conduct a competition on an unseen NLP task, winner will get a boost to their final grade

- **Student lectures**

You will have the opportunity to present papers to the class.

Lecturer Office Hours

Please coordinate in advance by email/phone.

Kfir Bar:

You may use both of the following addresses:

kfir.bar@post.idc.ac.il, barkfir@yahoo.com (expected response time is 24 hours, and no more than 48 hours).

Amir Cohen:

Amir.Cohen02@post.idc.ac.il (expected response time is 24 hours, and no more than 48 hours).

Tutor Office Hours

On demand

Teaching Assistant

Sahar Millis	sahar.millis@post.runi.ac.il
Royi Rassin	isroei5700@gmail.com

Reading List

Textbooks:

- Speech and Language Processing, 2nd Edition, by Daniel Jurafsky and James H. Martin [J&M]
- Foundations of Statistical Natural Language Processing, by Christopher Manning and Hinrich Schuetze [M&S]